

physiology

1- Which ions are exchanged between two sides of red blood cells in venous blood?



HCO_3^- and Cl^-

HCO_3^- and H^+

H_2CO_3 and HCO_3^-

H^+ and Cl^-

2- Which gas is used for estimating of "diffusing capacity of O_2 " in the lung?

Helium

CO_2



CO

O_2

3- What is the mean O_2 pressure difference (mmHg) across the respiratory membrane during normal and quiet breathing?

64

21



11

17

4- Which one the following is correct about ventilation and perfusion in the lungs?

The blow flow decreases from top of the lung to bottom



The ventilation increases from top of the lung to bottom

The ventilation to perfusion ratio (V/Q) increases from top of the lung to bottom

V/Q ratio is same for all levels of the lung in upright person

5- Compared to other gases, which gas has the highest diffusing capacity during exercise?

CO₂

O₂

CO

N₂

6- Which one of the following leads to an increase in airway resistance?

Vasoactive intestine Peptide (VIP)

Dynorphine

Norepinephrine



Cold air

7- What is the normal frequency for beating of cilia in mucociliary system?

10-20 beat per second

20-40 beat per minute

10-20 beat per minute

20-40 beat per second

8- Which one of the following happens during inspiration?

The antero-posterior dimension of the chest is decreased.

The diaphragm lifts up.

The ribs move downward.



The vertical dimension of the thorax is increased

9- Which of the following is NOT a form by which CO₂ can be transported in the blood?

Dissolved in the blood



As carboxyhemoglobin.

Bound to the amino end groups in hemoglobin

As bicarbonate

10- Which one the following factor does reduce intracellular oxygen consumption?

A decrease in intracellular ATP concentration

An increase in tissue blood flow



An increase in intracellular ATP concentration

An increase in intracellular ADP concentration

11- What is the normal value of utilization coefficient for O₂ in a normal person?

25%

20%

85%

75%

12- Which one the following leads to an increase in affinity of hemoglobin for oxygen?

- ☐ A decrease in pH
- ☒ A decrease in DPG concentration
- ☐ An increase in CO₂ pressure
- ☐ An increase in body temperature

13- Which one of the following is NOT correct about pulmonary circulation?

- ☒ Vasodilation occurs in response to hypoxia
- ☐ Supplies blood flow to respiratory zone
- ☐ Is a low-pressure circulation
- ☐ Provides blood flow for gas exchange

14- With a duration of 1.5 seconds for inspiration phase, what will be the duration of Ramp Signal?

- ☐ 2.5
- ☐ 3
- ☒ 1.5
- ☐ 2

15- What is the cause of "physiological shunt"?

Mixing of pulmonary vein blood with bronchial vein blood

A decrease in the cardiac output to half

An increase in VA/Q ratio 2.5 times more than normal amount

✓ A decrease in ventilation/perfusion ratio (V_A/Q) to zero

16- The terminal bronchioles are part of ...

Anatomical shunt

✓ Anatomical death space

Respiratory zone

Alveolar death space

17- Which of the following is correct about blood flow at the "top of the lung" in a healthy person while lying down?

Blood flow will only be maintained during systole

✓ There is a continuous blood flow during both systole and diastole

The blood flow is zone-1 type

The blood flow is intermittent

18- Which one of the following is correct about "Transpulmonary Pressure (TPP)"?

TPP= alveolar pressure minus atmospheric pressure

At the beginning of the inspiration is equal to + 5 cm of water

TPP= pleural pressure minus alveolar pressure

✓ At the beginning of exhalation is equal to – 7.5 cm of water

19- Which one the following is less than others in the lung?

Inspiratory Reserve Volume

Residual Volume

✓ Expiratory Reserve Volume

Inspiratory Capacity

20- Which one the following is correct about the role of the dorsal respiratory group in medulla?

Becomes active during deep inspiration

Inhibits the pneumotaxic center in pons

Becomes active during deep expiration

✓ Sends its action potentials to phrenic nerves

21- Which one the the following neurotransmitter is NOT released from glomus cell in carotid bodies?

Acetyl choline

Dopamine

✓ Noradrenaline

ATP

Histology

22- What Is The Name Cell Secreted Surfactant In Bronchiole?

- ☐ Simple Cells
- ☒ Celera
- ☐ Pnomocyte III
- ☐ Macrophage

23- Which of the following is observed in structure Histology In Respiratory Zone?

- ☐ Dense Connective Tissue
- ☒ Areolar Loss Connective Tissue + Erectile Tissue
- ☐ Stratified Squamous Epithelial Tissue
- ☐ Simple Cell

24- سوال Which part of the lung tissue has Nomocyte II?

- ☐ Bronchus
- ☐ Bronchiole
- ☒ Alveolar
- ☐ Terminal Bronchiole

25- In which of the following isn't observed

Goblet Cell ?

- ✓ Vestibule Zone
- Olfactory Zone
- Respiratory Zone
- Pharynx

26- سوال In which of the following is observed Elastic Fiber In Histology Structure ?

- Respiratory Zone
- Vestibule Zone
- ✓ Theracea
- Alveolar Duct

Anatomy

27- Which of the following paranasal sinuses open to the apex of bulla ethmoidal ?

- Anterior ethmoidal sinus
- Frontal
- ✓ Middl ethmoidal sinus
- Maxillary

28- All of the following bones participates in the formation of the lateral wall of the nose except?

- ✓ Vomer
- Maxilla
- Palatine
- Lacrima

29- All of the following structures are seen in inlet of larynx except?

- ✓ Vestibular fold
- Arythnoid cartilage
- Ary-epiglottic fold
- Epiglot cartilage

30- Which bone is involved in both the roof and the medial surface of the nasal cavity?

- Ethmoid
- Sphenoid
- ✓ Palatine
- Inferior concha nasalis

31- Which of the following is the highest structure in the right lung umbilicus?

- ☐ Pulmonary artery
- ☒ Superior upper lobe bronchus
- ☐ Pulmonary vein
- ☐ Main bronchus

32- The medial surface of the right lung is adjacent to all the following structures except

- ☐ Azygos vein
- ☐ Vena cava superior
- ☐ Vena cava inferior
- ☒ Arch of aorta

33- Which of the following arteries passes through the foramen in roof of nasal cavity?

- ☐ Superior labial artery
- ☐ Greater palatine artery
- ☐ Sphenopalatin artery
- ☒ Anterior ethmoidal artery

34- Which of the following nerve pass through incisive canal ?

- ☐ Posterior ethmoidal
- ☒ Nasopalatine
- ☐ Greater palatine
- ☐ Anterior ethmoidal

35- Which of the following muscles is involved in pulling thyroid cartilage downward and forward ?

- ☐ ...
- ☐ Oblique arytenoid
- ☒ Cricothyroid
- ☐ Posterior crico- arytenoid
- ☐ Transverse arytenoid

36- What is the name of the upper and free side of the quadrangular membrane?

- ☒ Ary -epiglottic fold
- ☐ Median thyrohyoid ligament
- ☐ Vesibular ligament
- ☐ Vocal ligament

37- All of the following muscles are paralyzed by cutting the recurrent laryngeal nerve except?

Thyroepiglottic

Lat Crico- arytenoid

✓ Cricothyroid

Posterior crico- arytenoid

Embryology

38- In which time of development the second epithelial alveolar cells appear?

3th month

4th month

5th month

✓ 6th month

39-

.In which period of lungs development alveolar duct appear

Pseudoglandular phase

Alveolar phase

Saccular phase

✓ Canalicular phase